A bound from the group generated by all panels

Human Compression project

Fix a finite group G and a subgroup B. Suppose that an acquired complete transport procedure supplies the actual list $\mathcal{P}_B$ of admissible panel subgroups above this fixed B. We consider only families of these panels that generate G. The completeness of the panel list, including its local action conditions, is an input to the following argument, not a conclusion of it.

Lemma 1. *Put $J_B = \langle B, P \mid P \in \mathcal{P}_B \rangle$. If $J_B < G$, no family under consideration exists at B. If every $P \in \mathcal{P}_B$ has index two over B, then $J_B \leq N_G(B)$, so a proper normalizer suffices for the same conclusion.*

Proof. Every selected panel lies in J_B. The group generated by any selected family therefore lies in J_B, and cannot equal G when $J_B < G$. In the second assertion, a subgroup of index two is normal. Thus each P normalizes B, and the same holds for the group they generate together with B. □

For the connected coset geometries in the acquired primitive augmentation theorem, generation of G by the panel subgroups is part of the established correspondence. The list $\mathcal{P}_B$ is obtained by the already proved transport of complete maximal inclusions; for the Higman–Sims source it is restricted to the inclusions with a two-transitive local action. The bound applies to this exact list. It makes no assertion that proper generated families are absent, and supplies no classification of them.

The implementation computes the same complete actual panel list as the accepted enumeration, forms J_B, and tests properness. An exclusion saves explicit generating permutations for B, every panel, and J_B. It has its own terminal status. It is not an empty final level of the old enumeration. If $J_B = G$, the wrapper invokes the unchanged old procedure. This may repeat the inexpensive construction of its panel list; it changes no recognition, equivalence or pruning rule inside that procedure.

For example, the retained Higman–Sims case numbered 96 has $|B| = 16$ and 19 admissible panels, all of order 32. Their generated group has order 512, strictly less than $|G| = 44\,352\,000$. The completed lower levels of an interrupted enumeration remain valid, but are unnecessary for this exclusion. The unfinished rank is not credited as exhausted. The exact subgroup classification and finite-group software premises used to obtain the panel list remain explicitly acquired.

This is an elementary bound used to avoid unnecessary enumeration. No novelty, open-conjecture, whole-source or compression claim is made by the argument alone.